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Predicting District Heating Networks Fault Location with Graph Neural Networks

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Abstract

District heating networks (DHNs) are critical infrastructure prone to physical failures such as leakage-related faults, which cause significant energy and financial losses. Traditional physics-based monitoring methods are computationally expensive and require the continual recalibration of complex mathematical models, while standard data-driven approaches often fail due to the scarcity of real-world sensor data. This study addresses these challenges by proposing a topology-aware graph neural network (GNN) architecture for fault localization. The methodology follows a two-stage process: first, a graph attention-based architecture is designed and optimized using a synthetic dataset to effectively capture multi-step neighborhood dependencies. Second, the model is adapted and evaluated on a physically simulated dataset of a real urban DHN, comprising 187 nodes and 42,570 operational states. The problem is formulated as a multi-class classification task across supply and return subnets. The results demonstrate high predictive performance, achieving an accuracy of 96% on the supply subnet and 91% on the return subnet. Analysis of prediction errors reveals a strong bias towards local topological mistakes, indicating the model's ability to capture the physical propagation of disturbances. These findings highlight the efficacy of GNNs in handling sparse data and exploiting network topology for robust DHN monitoring.

Keywords: district heating; graph neural networks; fault detection; fault localization; thermohydraulic simulation; network topology

1. Introduction

District heating is a vital component of urban infrastructure, supplying heat to dense urban populations worldwide. Like all infrastructure, it is susceptible to physical aging, corrosion, thermal stresses, and external impacts, leading to loss of insulation, pipe deformation, and reduced reliability of heat supply [1–3]. These changes disrupt the network's operation and ultimately result in failures. Such incidents cause not only substantial heat loss but also significant financial damage. Consequently, the detection and localization of faults in district heating pipelines is a key operational priority. This challenge, however, is not unique to district heating systems.

When examining fault detection, one observes a fundamentally similar class of diagnostic problems appearing across sectors such as gas [4], water [5], and power [6] distribution. However, oil/gas and water distribution networks have a significantly longer

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and broader industrial history. Consequently, many established fault detection methodologies and their practical applications have evolved primarily in these sectors, leaving district heating networks (DHNs) with less developed or adapted techniques [7].

From an engineering perspective, the probability and localization of DHN faults are also linked to load level, pipe-system design, material strength, insulation condition, operating regime, and weak points identified during pipe-static or fatigue assessment. These factors are normally addressed by national and regional standards. For the Russian DHN case considered in this study, the relevant standards include GOST R 55596-2013, GOST 30732-2006, GOST 25.101-83, and SNiP 41-02-2003 [8–11], while comparable European references include EN 13941 and EN 253 [12,13]. Some pre-insulated pipe systems also include built-in leakage-detection wires for local alarms when insulation becomes wet. The present study is not intended to replace such engineering and monitoring practices, but complements them by using sparse pressure and temperature measurements and network topology to rank likely fault locations when dense instrumentation or complete condition-history data are unavailable.

There are currently three main approaches to underground fault detection: physics-based methods [14], data-driven methods [15], and the analysis of infrared images captured by unmanned aerial vehicles (UAVs) [16].

Physics-based methods entail building a comprehensive, complex physical model, running numerical simulations, and comparing the model's predictions with real-world data (such as SCADA, or Supervisory Control and Data Acquisition, measurements). Differences between simulation results and reality offer limited options — effectively, the entire physical model must be reconstructed or recalibrated. This is a sufficiently complex and demanding process. Moreover, the model must either be intricate enough to account for external factors, corrosion, and measurement noise, or its continual recalibration will become a regular activity [17].

To avoid constantly updating physical models to evolving operational data, a direct utilization of such data (for instance, as multivariate time series) offers an alternative way for fault prediction. This approach gives rise to data-driven machine learning methods [18], which apply mathematical statistics to establish correlations between monitoring data and defect occurrences. Moreover, a wide range of modeling techniques is employed in this field, including gradient boosting [19], convolutional neural networks [20], physics-informed neural networks [21], each introducing distinct assumptions and inductive biases. Nevertheless, the effectiveness of these methods is highly dependent on data quality, particularly regarding sensor placement density, spatial distribution, and measurement frequency (with some approaches requiring the deployment of extra sensors or the collection of external measurements [22]).

However, applying these methods remains difficult. Models that demonstrate excellent performance during the development phase often prove non-transferable to real-world data. Data labeling inconsistencies, mismatched available features, a limited number of accessible parameters, and incomplete sensor coverage – all of these factors undermine the "data" component, causing data-driven methods to falter. The common modeling assumption of having complete information about the network where defects are to be localized leads to a dual outcome: on one hand, accuracy on synthetic data can be outstanding, yet on the other hand, it renders training such a model for a real-world DHN practically impossible [23–27].

A common strategy to address data scarcity involves a hybrid approach: generating synthetic data through physical simulation, followed by model training on this augmented dataset [28]. This methodology enables both the evaluation of various algorithms prior to deployment and the enhancement of existing models [29]. The underlying data shortage, however, remains both at the systemic level (a lack of public datasets and benchmarks) and at the physical level, given that sensor coverage across the network pipelines is usually poor (thus, creating the need for their smart placement [30,31]), severely limiting the ability to develop robust and generalizable models.

To maximize the utility of the limited data available, a promising approach is to look at the heating network's structure. The propagation of thermal and hydraulic transients within DHNs exhibits strong spatial locality: disturbances originating at a given location primarily influence immediately adjacent network elements before attenuating or dispersing further [32]. This neighborhood-dependent behavior establishes network topology as a primary determinant of transient dynamics and, consequently, of observable fault signatures.

Accordingly, a physically consistent modeling framework must explicitly preserve these spatial interdependencies. Representing the distribution system as a graph (where nodes correspond to junctions, consumers, or injection points, and edges represent connecting pipe segments) naturally encodes the adjacency relationships that govern wave propagation [24]. Within this representation, graph neural networks (GNNs) provide a theoretically grounded approach for fault analysis, as their message-passing mechanism inherently aggregates state information across topologically connected elements, mirroring the physical diffusion of thermal and hydraulic disturbances.

Recent studies on water and gas distribution systems show why graph-based learning is relevant for infrastructure diagnostics. Wu et al. [33] used GNNs for leakage-risk assessment in water distribution networks, while Zhang and Fink [34] proposed an algorithm-informed GNN for leakage detection and localization. Li and Wu [35] introduced a segment-feature fusion strategy for water-network leakage detection, Şahin and Yüce [36] reported leakage prediction results with graph convolutional networks, and Truong et al. [37] demonstrated GNN-based pressure reconstruction. For gas networks, Zhang et al. [38] investigated probabilistic GNNs for leak detection and localization. These studies indicate that topology-aware models can outperform topology-agnostic approaches because they exploit the structural constraints that govern how anomalies propagate through interconnected network components. However, most of this evidence comes from water/gas networks or denser sensing settings, leaving a gap in topology-aware DHN fault localization under sparse SCADA-like measurements and limited labeled field data. Inspired by these findings, this paper investigates the applicability of GNNs to DHNs, aiming to adapt these topology-aware architectures to the specific challenges of heat distribution monitoring.

2. Materials and Methods

2.1. Overall Approach

This research introduces a two-stage methodology designed to address data scarcity and complexity in DHNs. First, to decouple topological learning from physical modeling, the design and validation of the GNN architecture are performed using a fully abstract synthetic dataset (random trees). Subsequently, this architecture is tested and adapted on data generated by a physical simulation of a real DHN, also involving data preprocessing and performance evaluation.

2.2. Stage 1: Architecture Design on Synthetic Data

The objective of this stage is to investigate and design a GNN architecture capable of effectively capturing and aggregating information from multi-step (k-hop) neighborhoods of graph edges. Since real DHN data is limited, the architecture must demonstrate this capability with minimal computational complexity and without an excessive number of parameters.

To achieve this, we generated an artificial dataset, built and conducted experiments with various neural network models, and examined how different hyperparameters and mechanisms (convolutions, attention, pooling, skip-connections, etc.) affected the performance of these models. The target variable for each edge was explicitly defined as a function of the features of nodes located within k steps from that edge (see equation 1), which simulates physical dependencies in networks. This simple and well-defined synthetic dataset was created to enable a fully controlled experiment for architecture selection.

To formulate a regression task on graph edges, we designed a dataset as a set of random trees. The node features in a tree included a random scalar value and the two-dimensional coordinates (x, y) of the node in an arbitrary planar embedding of the graph on the plane. The edge features consisted only of random scalar values. The target variable for each edge e was defined as a function $f_k(e)$:

$$f_k(e) = \sum_{i:w_i \in N_k(e)} X_i \cdot \lambda(w_i, e), \tag{1}$$

where $N_k(e)$ is the set of nodes within a k-hop neighborhood of edge e, $X_i \in [0, 1]$ is the node attribute, and λ is a decay factor.

We tested several different decay factors, including:

- $\lambda = 1$ – constant decay,
- $\lambda = 2^{-d}$, $d = \arg\min_k \{w_i \in N_k(e)\}$ – distance-based exponential decay,
- $\lambda = \prod_{w_i \in P} W_i$, where P is the set of all edges in the shortest path between edge w_i and edge e (excluding e), and $W_i \in [0.1, 1]$ is edge feature of w_i – random exponential decay of the node feature weight based on the distance.

An example with $\lambda = 1$ is illustrated in Figure 1.

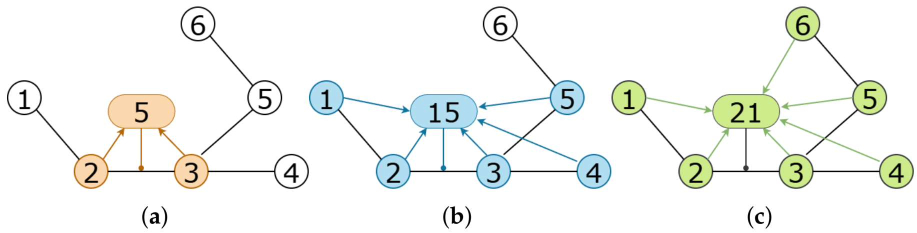


Figure 1. Illustration of target variable behavior across different k: (**a**) $k = 0$. (**b**) $k = 1$. (**c**) $k = 2$.

2.3. *Proposed Architecture.*

Experiments were conducted with various GNN architectures, including different kinds of message-passing, GCN, GAT, and GraphSAGE [39–41] layers, as well as regularization methods such as Dropout, LayerNorm, and GraphNorm. Ultimately, an architecture utilizing an attention mechanism was proposed, demonstrating strong performance. The constructed GNN consists of two main components:

1. A **node encoder** using a sequence of GATv2 layers, LayerNorm, and ReLU activation, aggregated via Jumping Knowledge with concatenation [42,43].
2. An **edge attention layer** implementing multi-head attention over edges sharing a common node, followed by LayerNorm and skip-connections.

The main findings from the architecture search and ablation study were: first, the use of spatial node coordinates (x, y) significantly improved prediction quality; second, the attention mechanism was a key component regardless of the complexity of the task;

and third, the selected architecture outperformed all others considered in learning local functional dependencies.

2.4. Physically Modeled Dataset Generation

To generate the dataset required for training the GNN, a thermohydraulic simulation was employed to model the physical behavior of the DHN under various operational and fault conditions. The heating network in the simulation is represented as an oriented graph, where the nodes correspond to pipe junctions and heat sources, and the edges represent pipeline branches with specified geometric, thermal, and hydraulic parameters. Notably, to close the hydraulic circuit, the end consumers are modeled as specialized edges connecting the supply and return sub-networks. These consumer edges are assigned specific local hydraulic resistance coefficients and heat transfer parameters to simulate the pressure drop and thermal extraction of building's internal heating system. The flow distribution within the network is governed by the hydraulic circuit theory, requiring the simultaneous solution of mass and momentum conservation equations [44].

The hydraulic state is determined from conservation of mass at each node and a momentum (head-loss) relation on each branch. Let branch $b = (i, j)$ connect nodes i and j, and let its positive direction be chosen from i to j. The unknown hydraulic variables are the nodal pressures p_i and the branch mass flow rates m_b, whose sign indicates the flow direction relative to the branch orientation. In discrete graph form, nodal continuity is written as

$$\sum_{b \in \mathcal{U}_i} \varepsilon_{ib} m_b = q_i, \tag{2}$$

where $\mathcal{U}_i$ is the set of branches incident to node i, $\varepsilon_{ib} = \pm 1$ is the incidence sign determined by branch orientation, and q_i represents the external mass source or sink at the node.

The momentum conservation along each pipeline segment is described by the nonlinear relation

$$s_b |m_b| m_b = -\Delta p_b, \tag{3}$$

where $\Delta p_b = p_j - p_i$ is the pressure drop across branch b, and s_b is the effective hydraulic resistance coefficient of the branch. In general, s_b is determined by the hydraulic properties of the branch and may include both distributed friction and local hydraulic losses. In this case, the effective resistance can be written as

$$s_b = \frac{1}{2\rho_b A_b^2} \left(\lambda_b \frac{L_b}{D_b} + \xi_b \right), \tag{4}$$

where λ_b is the Darcy friction factor, L_b is the branch length, D_b is the hydraulic diameter, A_b is the cross-sectional area, and ξ_b is the sum of local hydraulic resistance coefficients. In the present simulations, the Darcy friction factor was evaluated using the explicit Altshul correlation,

$$\lambda_b = 0.11 \left(\frac{\delta_b}{D_b} + \frac{68}{Re_b} \right)^{0.25}, \tag{5}$$

where δ_b is the equivalent roughness of the pipe wall, μ_b is the dynamic viscosity, and

$$Re_b = \frac{\rho_b |u_b| D_b}{\mu_b} = \frac{|m_b| D_b}{\mu_b A_b} \tag{6}$$

is the branch Reynolds number. This explicit relation was used to avoid an additional inner iteration for λ_b within the global hydraulic pressure-correction procedure.

To solve these equations, the simulation employs a SIMPLE-like iterative procedure. This algorithm utilizes an explicit equation for pressure correction,

$$\nabla \cdot (\kappa \nabla p') = \nabla \cdot m - q, \tag{7}$$

where p' is the pressure correction and

$$\kappa_b = \frac{1}{s_b |m_b|}. \tag{8}$$

The solver sequentially updates the nodal pressure field and branch mass flow rates through iterations until strict mass-balance convergence is achieved across the entire network.

Thermal calculations are coupled to the hydraulic solution by evaluating convective heat transport and thermal losses to the environment after hydraulic calculation iteration. Heat loss from a pipe to ambient air is modeled by the Newton–Richmann relation

$$Q = \alpha F (t_b - t_{out}), \tag{9}$$

where α is the overall heat transfer coefficient, F is the pipe surface area, t_b is the average water temperature within the pipe, and t_{out} is the ambient outdoor temperature. The average pipe temperature t_b is calculated from the steady one-dimensional energy balance with distributed heat loss,

$$t_b = t_{out} + \frac{\dot{m}_b c_p}{\alpha F}(t_1 - t_{out}) \left[1 - \exp\left(-\frac{\alpha F}{\dot{m}_b c_p} \right) \right], \tag{10}$$

where $\dot{m}_b$ is the branch mass flow rate, c_p is the specific heat capacity, and t_1 is the inlet temperature of the pipe.

Equation (10) describes a steady branch-integrated solution of the one-dimensional energy balance with distributed heat loss. Under the quasi-stationary assumptions used in this work, the logarithmic mean thermal driving force is applicable over a branch of arbitrary length, provided that the mass flow rate, heat-transfer coefficient, ambient temperature, and thermophysical properties are treated as constant or branch-averaged within that branch. The branch length is therefore already accounted for through the heat-transfer surface area F and the exponential attenuation factor $\alpha F / (\dot{m}_b c_p)$, additional axial subdivision is not required to evaluate the steady heat loss of a uniform branch. This formulation, however, does not resolve thermal transport delays, hot-water-front propagation, or pipe-wall thermal inertia. These effects are outside the scope of the present dataset, where each sample represents a quasi-stationary operating regime used for fault localization, and the sequential evolution of hydraulic and thermal states during a transient process is not modeled.

At network nodes where multiple fluid streams intersect, perfect adiabatic mixing is assumed, and the resulting nodal temperature is computed as the mass-weighted average of the incoming flows. All thermal properties of the heat carrier fluid, including density, dynamic viscosity, and specific heat capacity, are dynamically updated during the iterations based on the local fluid temperature.

To unify the hydraulic and thermal calculations, an iterative steady-state simulation algorithm is used. The computational sequence starts by assigning initial approximations of nodal pressures and temperatures from the network boundary conditions. Based on these values, temperature-dependent thermophysical properties of the fluid are evaluated and initial branch mass flow rates are estimated. The core of the algorithm then enters an iterative hydraulic loop: the branch mass flows are updated from an iterative solution of the momentum conservation equations, after which a global pressure-correction equation is solved to adjust the nodal pressure field and correct the branch flows.

Once the hydraulic flow distribution is stabilized, the algorithm proceeds to the thermal step and updates the temperature field. The fluid density, viscosity, and heat capacity are then recalculated for each pipe using the newly obtained local temperatures. The coupled iterations continue until the predefined convergence criterion is satisfied; otherwise, the solver returns to the hydraulic phase with updated thermophysical properties.

Using this simulation approach, a local segment of an urban DHN located in Arkhangelsk Oblast, Russia, was modeled. The case study represents a compact residential district served by a central heat source and includes consumer stations together with the interconnecting supply and return pipeline infrastructure. The resulting graph contains 187 nodes and 202 pipe segments. Thus, the modeled object is a local hot-water supply/return DHN segment rather than a complete city-wide heat-supply system. In this model consumer stations are heat sinks, and the source/sink symbols in Figure 2 denote the heat-source supply/return boundary nodes. The heat source is represented by boundary conditions: specified supply flow rate and temperature, and fixed return pressure, while the boiler outlet pressure is obtained within the hydraulic solution.

To provide a representative range of operating conditions, simulations followed an operational temperature curve that maps outdoor ambient temperature to the required source supply temperature and mass flow rate. Consumer heat demands were also scaled according to the same curve. The outdoor temperature was varied from $-34°$C to $+10°$C with 1°C increments to define baseline environmental scenarios.

Therefore, the boundary conditions in the generated dataset are scenario-dependent values defined by this operating curve, where the ambient-temperature range specifies the load envelope covered by the simulations.

For each baseline scenario, a grid of operational states with various modifications of baseline scenario, was generated, resulting in a final dataset of 42570 samples. Natural data noise (environmental fluctuations and measurement uncertainty) was simulated by applying random variations within 5% of heat transfer coefficient α on a randomly selected 10–20% subset of pipes. To emulate localized thermal degradation (e.g., insulation failure or pipe flooding), a set of defect scenarios in which α was increased in a single pipeline segment was generated. The defect location was swept over all pipeline segments in the network, excluding consumer-modeling pipes. Five defect severity levels were considered, corresponding to increased heat transfer in the ranges 120–150%, 150–170%, 170–200%, 200–300%, and 300–400% of the baseline α value.

Consumer-modeling pipes are auxiliary graph edges that connect the supply and return subnets through a consumer station and represent consumer-side hydraulic resistance and heat extraction. They are not treated as candidate distribution-pipeline fault classes in the present localization task.

In this formulation, a defect is not modeled as a mass leak: no water-loss term is added to the nodal mass-balance equation and the branch mass-flow equations remain closed. The fault corresponds to a local increase in heat transfer through the pipe wall/insulation. A true leakage case with water loss would require an additional mass source/sink term in the continuity equation and changed branch flows; this case is outside the scope of the current dataset.

The dataset generation process produced the following samples for each temperature regime: 10 defect-free operational states and 935 defective states (5 simulations for each of the 187 pipes). Additionally, for each temperature, a reference sample without defects or noise was generated as a baseline. In total, the dataset contains 42,525 training samples (42,075 defective and 450 defect-free) plus 45 reference samples.

When preparing the data for training, the full graph was split into supply and return subnets, resulting in 85,050 total samples: 42,075 with a defect in the considered subnet (49.5%) and 42,975 without a defect (50.5%) - the latter includes 42,075 samples from the opposite subnet plus 450 defect-free samples from each subnet. This balanced distribution between defective and non-defective samples was intentionally designed to ensure robust model training.

2.5. Stage 2: Application to DHN Fault Localization

Stage 2 builds on the design choices and assumptions that worked well in Stage 1. In Stage 1, the objective was to verify the model's capacity to aggregate information from multi-step (k-hop) neighborhoods by learning functional dependencies from synthetic data (Equation 1). This capability is critical for DHN fault localization because physical disturbances, such as pressure drops or temperature anomalies, propagate spatially through the network. Consequently, the model must capture these local topological dependencies to accurately trace the fault source.

Having understood what a suitable GNN should look like, we moved on to fault localization in the DHN. To mimic realistic sparse monitoring, dynamic pressure and temperature values are retained only at 30 nodes: the heat-source supply and return nodes, and the supply/return connection points of the 14 consumers (2 nodes per consumer). This configuration mirrors real-world conditions where sensor coverage is sparse. Since edge-level regression is impractical under such sparsity, we reformulated the problem as multi-class classification. For this purpose, we adapted the architecture by replacing the final regression layer with a classification head and introducing global pooling.

We used a prepared dataset of 42,570 simulated operational states of a DHN, represented as a graph with 187 nodes (junctions, consumers, source/sink) and 202 edges (pipes) illustrated in Figure 2.

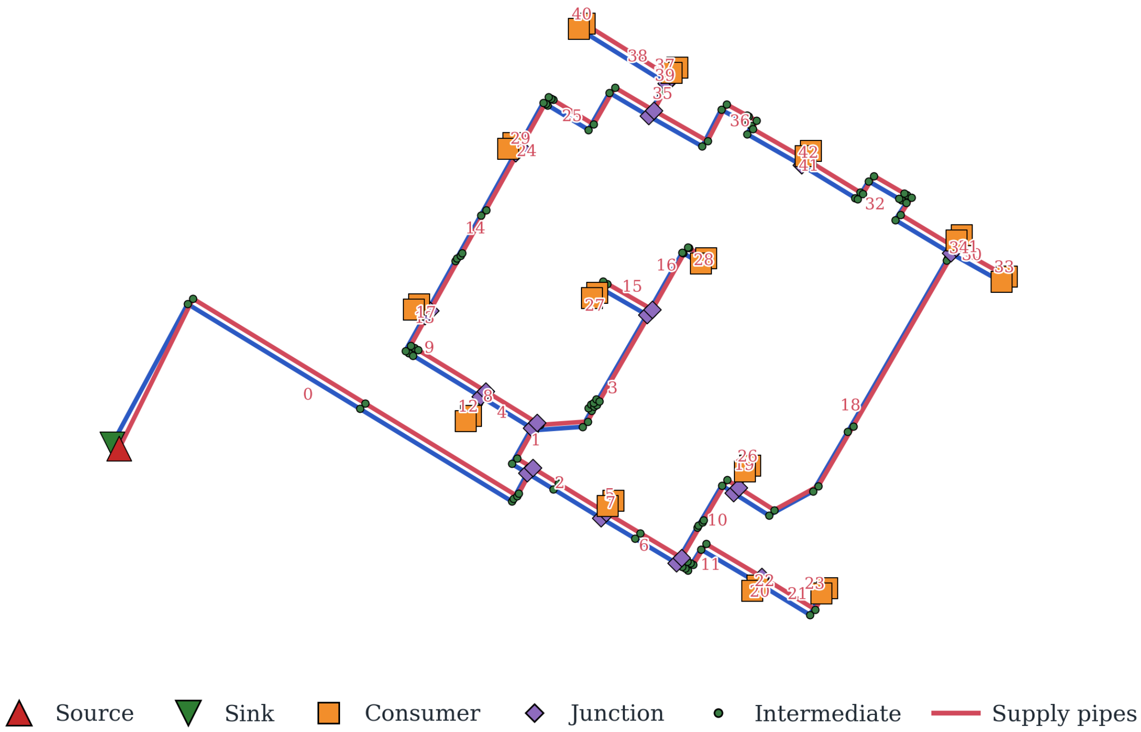


Figure 2. An illustration of the district heating network.

The following node features were used for model training:

- Local x, y coordinates of the node.
- Node type – source, sink, or pipe junction.
- Pressure at the node – a dynamic parameter.
- Temperature at the node – a dynamic parameter.

The following pipe (edge) features were used:

- Pipe diameter.
- Pipe length.
- Pipe type – supply, return, or consumer line.

Additionally, the outdoor air temperature was used as an input parameter.

The data preprocessing pipeline consists of the following steps:

1. Splitting the large heating network graph into two intersecting subgraphs – the supply and return subnets. These are formed by selecting all supply pipes, return pipes, and consumer pipes along with their incident vertices in the graph. These components have similar structures, as supply and return pipes are physically co-located.
2. Loading the data required for training, including global parameters such as outdoor air temperature and boiler plant temperatures, which are duplicated into separate tables.
   2.1 A numerical identifier is added to each edge in the graph, indicating which graph section the edge belongs to after partitioning. For each graph, a class (numerical label) is assigned – the index of the section containing the defective edge (or a separate class if no defects are present).
   2.2 The partitioning logic is as follows: edges belong to the same section if and only if they form a direct path between two key vertices. A vertex is considered a key vertex if it is a consumer vertex (incident to consumer pipes), a boiler plant vertex (heat source or sink), or a junction vertex (with in-degree + out-degree > 2).
   2.3 Adding dynamic parameters (readings from temperature and pressure sensors) to the graph vertices corresponding to consumers and the boiler plant (30 out of 187 vertices in total).
3. Transforming the raw data into a set of tabular representations.
4. Estimating normalization parameters and scaling three groups of features (the two heating network components and the global parameters) to zero mean and unit standard deviation based on the training set.
5. Applying normalization to all tabular data.
6. Converting the normalized tabular data into graph objects used by the model.

The core architecture from Stage 1 was adapted for classification as illustrated in Figure 3:

- The final edge feature representations, enhanced by the node encoder and edge attention blocks, were concatenated with their transformed versions from all attention layers.
- A global pooling layer aggregated edge features into a single graph-level representation.
- A linear classifier layer mapped this representation to logits over the classes.

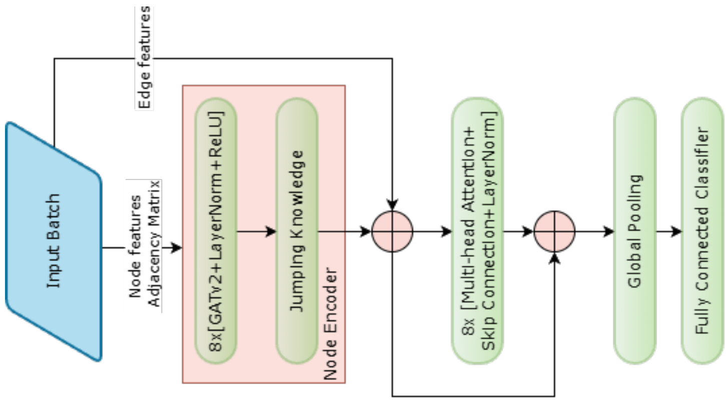


Figure 3. An illustration of the architecture of our GNN.

2.6. Training.

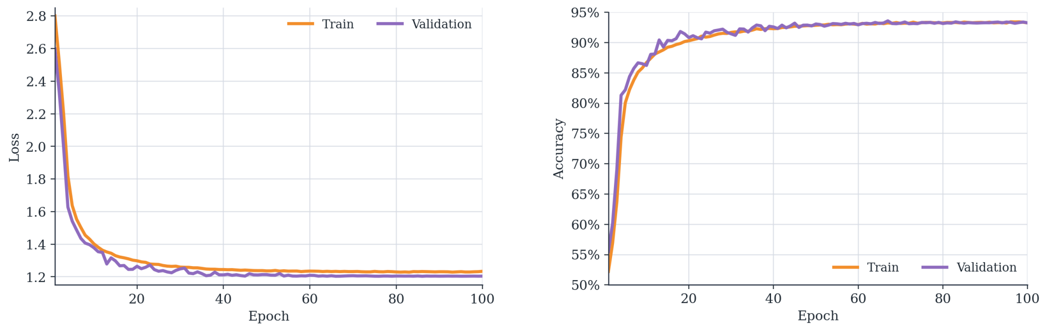
The model was trained for 100 epochs with a batch size of 16 graph pairs. The dataset was split into training, validation, and test sets with a ratio of 70:15:15. We conducted 3-fold cross-validation, obtaining consistent accuracy scores (0.913±0.011, averaged between two subnets) across folds, confirming that the results do not depend on a particular data split. The split was performed randomly over generated operational states; therefore, outdoor-temperature regimes and severity ranges may be represented in all subsets. The evaluation thus measures interpolation across fault locations and operating states within the simulated operating envelope, rather than extrapolation to completely unseen climate regimes.

The model was optimized using Adam [45]. The initial learning rate was set to 10^{-3}, with $\beta = (0.9, 0.99)$, $\varepsilon = 10^{-8}$, and weight decay $\lambda = 10^{-6}$. The learning rate was reduced exponentially each epoch down to 10^{-6} using a StepLR scheduler with a step size of 1 and $\gamma = (10^{-6}/10^{-3})^{1/100} \approx 0.933254$.

The loss function was multi-class cross-entropy (CrossEntropyLoss) with label smoothing ($\varepsilon = 0.1$) and class weighting applied. The model hyperparameters are listed in Table 1.

Parameter	Value
Node encoder layer dimension	128
Edge attention layer dimension	128
Number of node encoder layers	8
Number of edge attention layers	8
Number of attention heads	8
Jumping knowledge mode	`concatenation`

Table 1. Model Hyperparameters.



(**a**) Cross-entropy loss (**b**) Accuracy

Figure 4. Training vs. validation performance over epochs.

2.7. Implementation

All models were implemented using Python 3.12, PyTorch 2.4.1, and the PyTorch Geometric 2.6.1 library.

2.8. Baseline models

We implemented three non-graph baseline models: Random Forest, Histogram-based Gradient Boosting, and a Multi-Layer Perceptron. For each sample in the dataset, we extracted a feature vector comprising all available node-level (pressures, temperatures, coordinates, types) and edge-level (diameters, lengths, types) attributes, along with the outdoor air temperature, while disregarding the structural connectivity of the heating network. The classification target was the same as for the GNN: the index of the graph section containing the defective pipe (or a separate class for defect-free operation). All baselines were trained and evaluated using the same data split and the same random seed to ensure a fair comparison. Hyperparameters were kept at their default scikit-learn configurations, as preliminary tuning showed no substantial improvement over the defaults.

3. Results and Discussion

3.1. Main Results

The performance of the proposed GNN architecture was evaluated using the physically modeled dataset described in Section 2.5. To ensure a robust assessment of generalization, the model was trained using the Adam optimizer, with convergence observed after approximately 50 epochs as indicated by the stabilization of the validation loss on Figure 4.

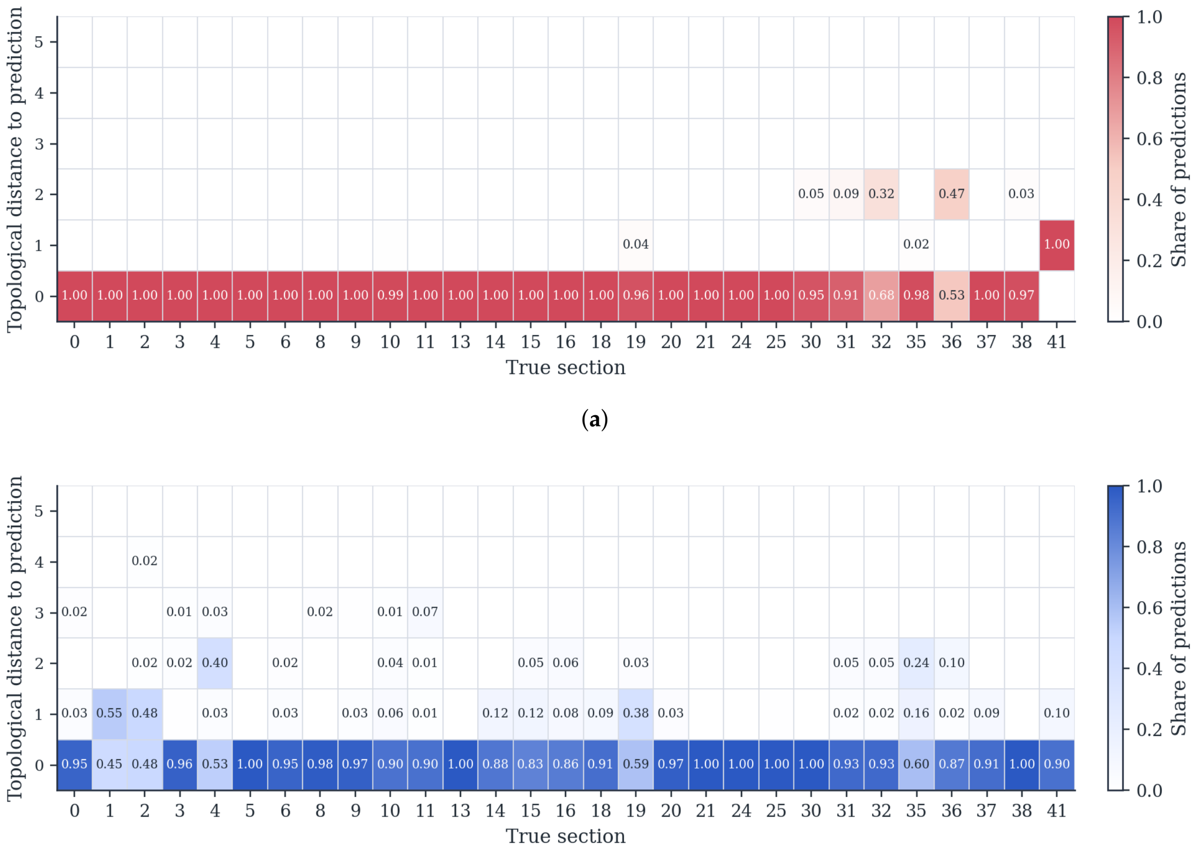
Table 2 presents the results achieved by our model on the test dataset. The reported number of classes (30) includes both the specific fault locations across the network sections and an additional class representing the normal operating state ("no defect"). The mean Accuracy and Macro F1 scores exceeding 90% demonstrate that the model exhibits strong predictive performance on sparse data, successfully capturing graph dependencies and effectively handling the classification task. For a given class, precision is the fraction of predictions assigned to the class that are correct, while recall is the fraction of true samples of the class that are detected. The F1-score is their harmonic mean, $F1 = 2 \cdot (\text{precision} \cdot \text{recall})/(\text{precision} + \text{recall})$. Macro F1 is the unweighted average of per-class F1 values and is therefore sensitive to performance on rare classes instead of being dominated by larger classes.

Model	**Supply Subnet**		**Return Subnet**	
	Accuracy	**Macro F1**	**Accuracy**	**Macro F1**
		Baseline models		
Random Forest	0.828	0.567	0.623	0.220
Hist. Gradient Boosting	0.829	0.591	0.675	0.379
MLP	0.867	0.711	0.646	0.266
		Proposed topology-aware model		
GNN	**0.960**	**0.930**	**0.910**	**0.869**
Number of classes	30		30	
Number of samples	6380		6380	

Table 2. Performance comparison between baseline models and the proposed GNN on the fault localization task. The GNN substantially outperforms all topology-agnostic baselines, particularly on the return subnet and the Macro F1 metric.

The observed performance discrepancy between the supply (0.960) and return (0.910) subnets aligns with the physical characteristics of the system. The return pipeline has a narrower range of dynamic values (temperature/pressure). Since the fluid has already cooled and pressure has dropped, the relative impact of a defect is smaller compared to the background noise.

To interpret the model's errors, we aimed to understand how far the predictions deviate from the ground truth when the model misclassifies (in terms of graph distance). For this purpose, we introduce a "distance confusion matrix" in Figure 5. The x-axis represents the true fault location (graph section), while the y-axis shows the topological distance between the predicted and true sections. Each cell indicates the proportion of predictions falling at a given distance from the true class. Perfect predictions would yield a horizontal line at distance 0 with value 1.0.



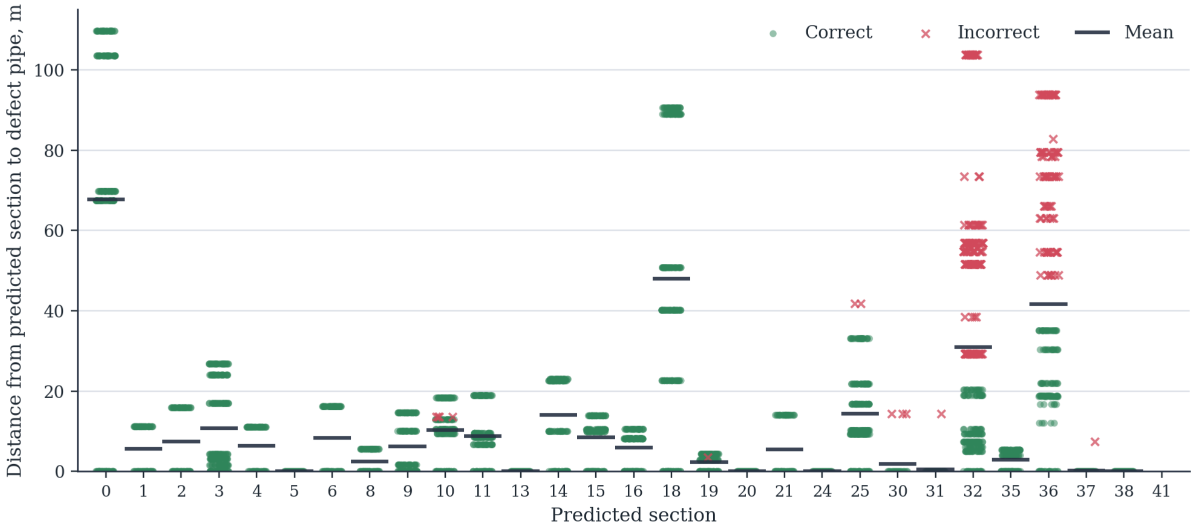
(**b**)

Figure 5. Distance confusion matrices: (**a**) Supply subnet. (**b**) Return subnet.

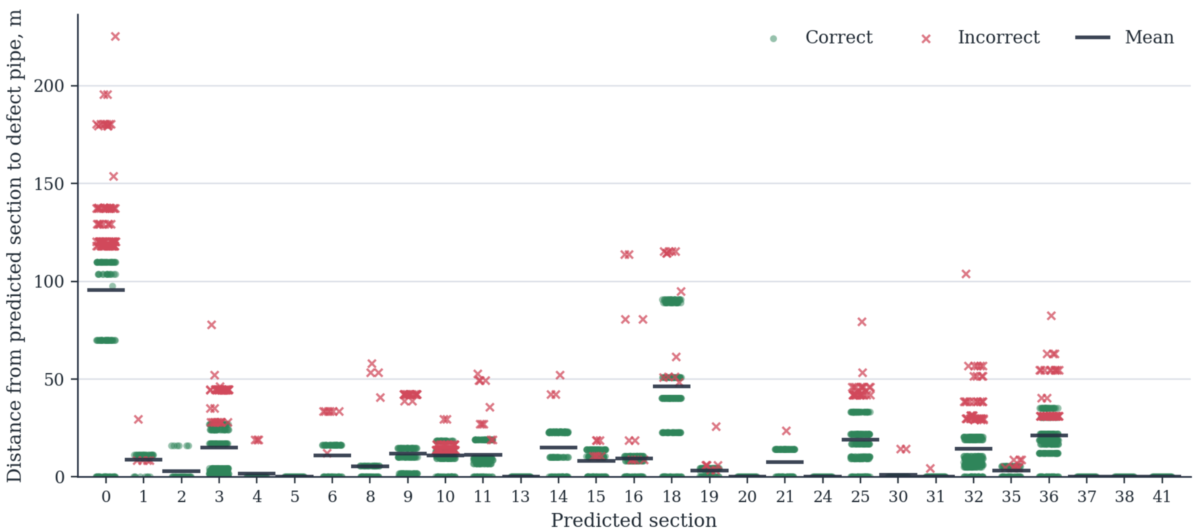
For segment 41, the value 1.0 at nonzero distance means that all 43 test samples belonging to this segment were assigned to another section. Segment 41 consists of a single pipe connecting a consumer to a branching node, resulting in fewer training examples compared to segments containing multiple pipes (N pipes × 225 defective samples per pipe). Additionally, segment 41 is topologically the farthest from the heat source among supply-side segments. Because of this, temperature and pressure signals at this segment are attenuated, making the defect less pronounced.. Consequently, the model misclassifies faults from segment 41 as belonging to neighbouring segments 32 or 36, each of which comprises more pipes. The confusion matrix therefore indicates a local ambiguity among segments 32, 36, and 41 rather than random class assignment errors. These sections are located in the most remote part of the supply subnet relative to the heat source, where pressure and temperature disturbances are weaker. This remoteness explains why defects in these segments are harder for the model to distinguish.

Notably, the distribution of errors in Figure 5 demonstrates a strong bias towards local mistakes. We observe that the error mass is heavily concentrated within a topological distance of 1–2 sections, with negligible occurrences of errors beyond that distance (relative to the network diameter of 12). Because heat loss and pressure drop affect neighboring segments, the model correctly identifies the neighborhood of the fault even if it misassigns the specific pipe edge. From an operational standpoint, this suggests that even in cases of misclassification, the model effectively narrows the search area to the vicinity of the actual fault. In practical terms, this means that the initial inspection area can be limited to the predicted section and its one- to two-section topological neighbourhood, before using additional field diagnostics to identify the exact pipe edge.

To complement the hop-based analysis, we also evaluated the metric localization error along the pipe network. For each prediction, the representative predicted position was taken as the midpoint of the predicted graph section, and the true position was taken as the midpoint of the defective pipe. The distance was then computed along the shortest path in the network using pipe lengths as edge weights. Figure 6 shows that the physical-distance analysis is consistent with the topological-distance result: most correct and near-correct predictions remain spatially close to the true defect, while the largest errors are concentrated in the same problematic regions identified by the confusion matrices.

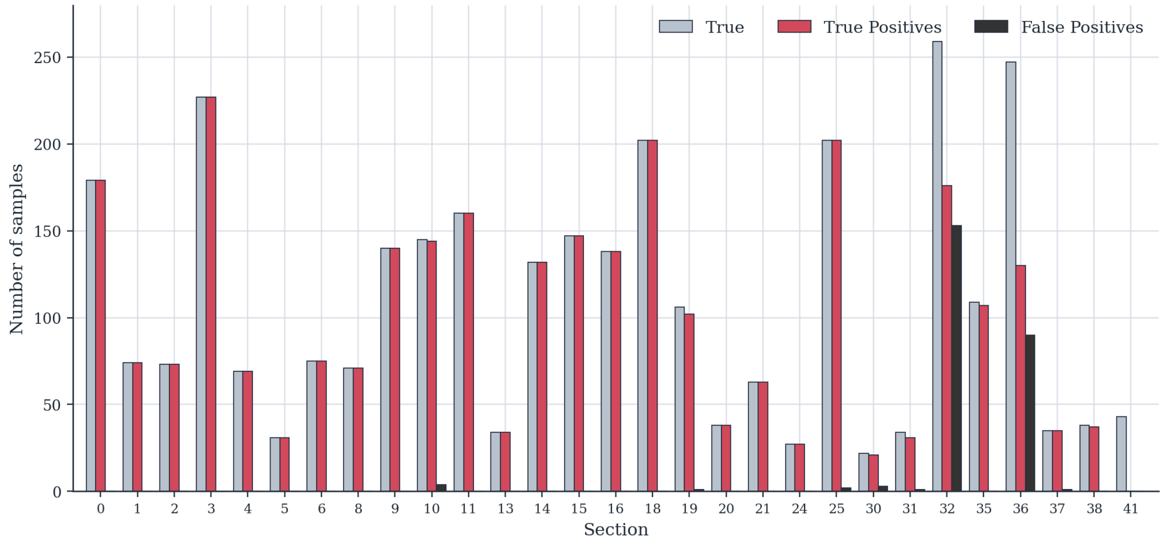


(**a**)

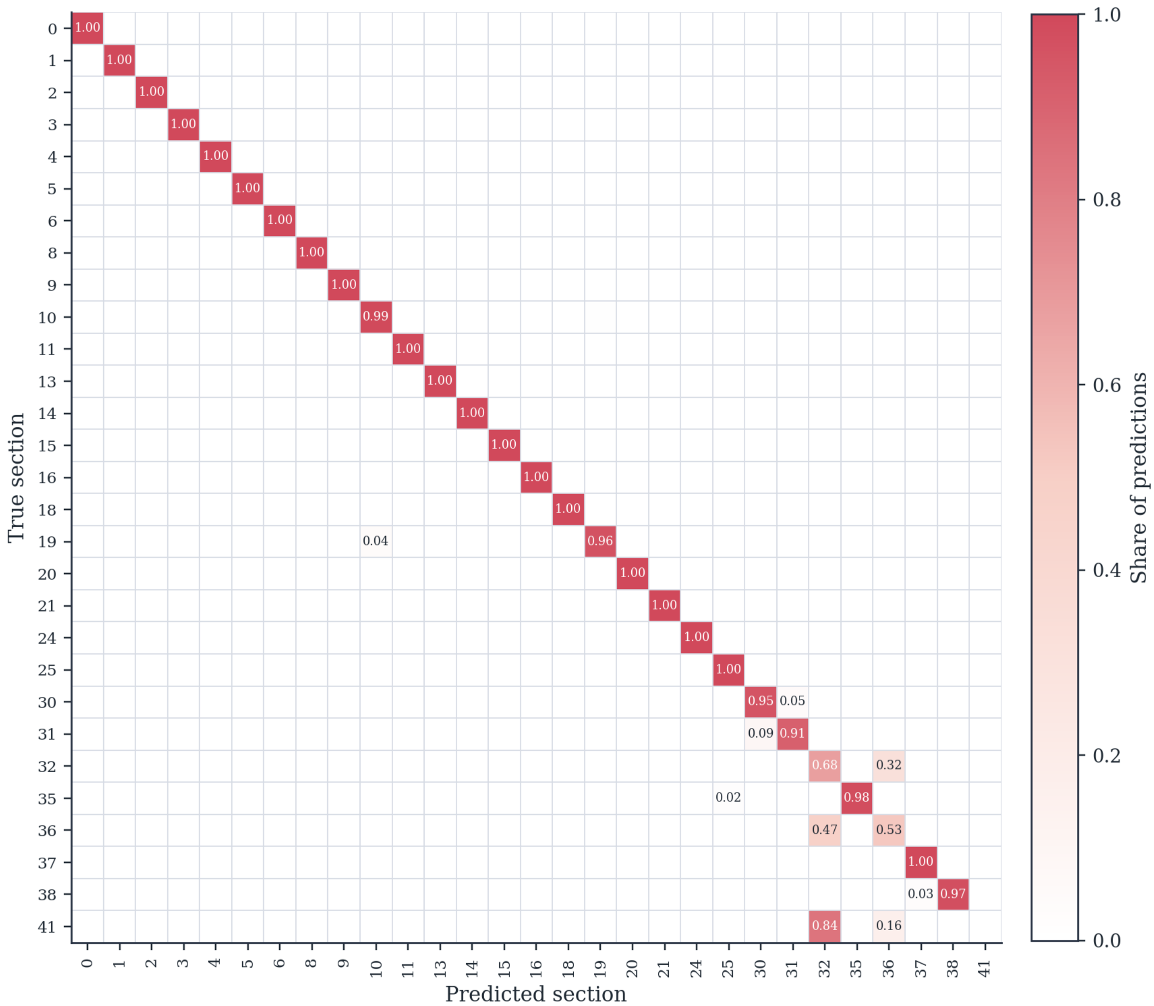


(**b**)

Figure 6. Metric localization error measured along the pipe network: (**a**) Supply subnet. (**b**) Return subnet. Green points correspond to correctly classified samples, red crosses correspond to misclassified samples, and the dark horizontal marks show the mean distance for each predicted section.

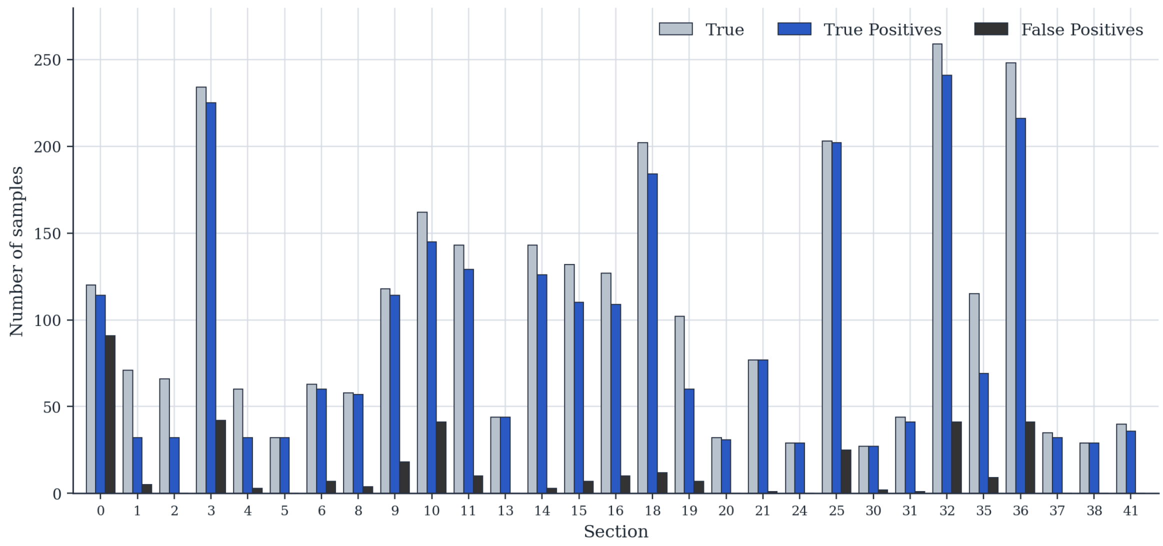


(**a**) Class distribution

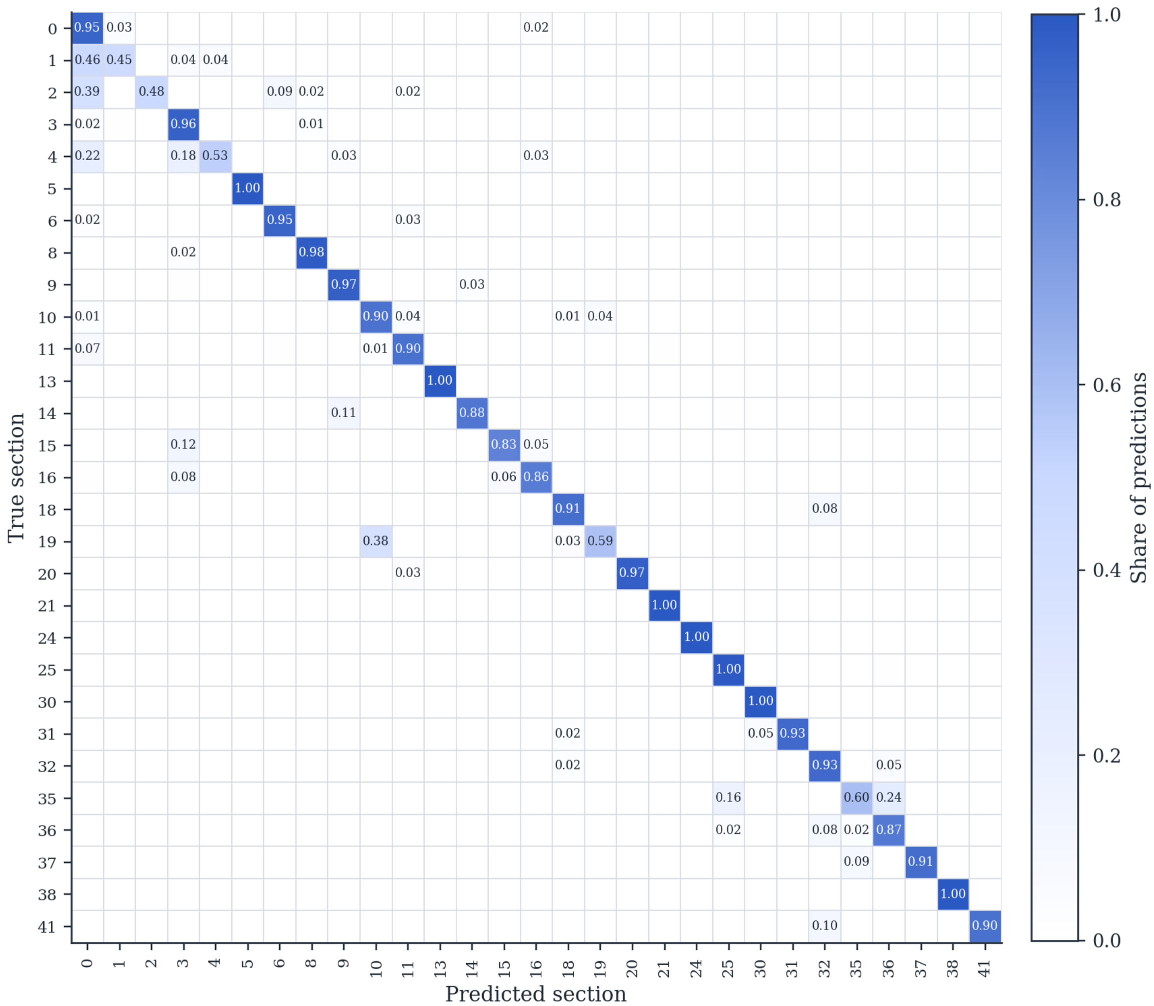


(**b**) Normalized confusion matrix

Figure 7. Supply-subnet diagnostics, excluding the "no defect" class: (**a**) distribution of predicted classes and (**b**) normalized confusion matrix.



(**a**) Class distribution



(**b**) Normalized confusion matrix

Figure 8. Return-subnet diagnostics, excluding the "no defect" class: (**a**) distribution of predicted classes and (**b**) normalized confusion matrix.

The class-distribution panels in Figures 7 and 8 show, for each network section, the number of true defect instances together with the counts of correctly and incorrectly predicted assignments. In these panels, "True" denotes the number of test samples whose simulated defect is located in the corresponding section, while "True Positives" and "False Positives" denote correct and incorrect predictions assigned to that section, respectively. Similar true and predicted counts can still hide compensating errors between neighbouring sections. Therefore, the distribution plots are considered together with the corresponding confusion matrices, which show the actual class-to-class error structure.

The distribution of predicted classes and the corresponding confusion matrices in Figures 7 and 8 support the earlier observations. For the supply subnet, the main errors are concentrated in the remote group of segments 32, 36, and 41, with segment 41 being the most problematic case; this confirms that the model has difficulty separating weak fault signatures in the farthest part of the network. A secondary supply-side ambiguity is observed for segment 35: defects in this segment are sometimes assigned to the adjacent and larger segments 25 and 36, which can produce similar aggregated hydraulic and thermal signatures while also being represented by more training samples. The asymmetry between the supply and return predictions for segment 41 can be explained by the flow direction and sensor placement. In the supply subnet, the sensor node at the end of segment 41 is the only sensor that lies downstream of long T-shaped group of segments 32, 36, and 41, while it may be the most sensitive to temperature disturbances originating in these segments, having just this single downstream reading may make it difficult for the model to pinpoint exactly which of the three segments contains the defect. In the return subnet, the flow direction is reversed. Sensors located downstream along the return path can therefore detect disturbances originating in segments 32 and 36 more explicitly than in the supply subnet, where only the segment 41 sensor lies downstream of those segments. This additional information may help the model separate these fault classes. It is worth noting that this particular arrangement of segments and sensors is unique to segment 41 within our network topology, which motivates the above explanation for the errors observed on this segment.

For the return subnet, the diffuse, scattered pattern of misclassifications across multiple classes confirms that the narrower dynamic range makes fault signatures less discriminative, and the noise begins to mask the underlying data patterns. In particular, return-subnet segment 0 is often confused with nearby segments 1, 2, and 4. This segment is closest to the return-side sink and therefore lies among the most distant sections from the source along the return path. As the distance from the source increases, hydraulic and thermal responses become smoother and defect signatures affect the measured parameters less explicitly. Moreover, segment 0 contains several long pipes, so disturbances originating in adjacent sections can produce similar aggregate signatures and be assigned to segment 0.

3.2. Ablation Studies

To validate our architectural choices, we conducted a comprehensive ablation study examining three key aspects: (1) model depth and the impact of multi-head attention mechanisms with skip connections, (2) pooling strategies, (3) number of attention heads, and (4) choice of graph convolution layer type. All experiments were performed on the fault classification task using the DHN dataset described in Section 2.5.

As can be seen in Tables 3, 4, the ablation experiments confirm our hypothesis that the proposed architecture is optimal among comparable alternatives, with the attention mechanism serving as a key factor in prediction quality, both in the node encoder and as an edge-attention layer.

Configuration	S (4 layers)		M (8 layers)		L (16 layers)	
	Acc	F1	Acc	F1	Acc	F1
Attention + skip connections	91.4	86.8	**92.1**	**87.6**	73.5	30.0
Attention w/o skip connections	88.7	79.2	**90.2**	**75.7**	73.5	31.5
No attention (MLP)	84.9	67.3	**86.6**	**76.5**	71.3	24.8
Number of parameters	725K		1.45M		2.9M	

Table 3. Ablation study results: impact of model depth, multi-head attention mechanism, and skip connections on classification accuracy (%) and macro F1-score (%).

Configuration	Accuracy (%)	Macro F1 (%)
Baseline: 8 layers, GATv2, attention + skip, 4 heads, attention pooling		
Pooling strategies		
Attention pooling	**92.1**	**87.6**
Mean pooling	86.7	65.9
Attention heads		
2 heads	88.7	78.5
4 heads	92.1	87.6
8 heads	**93.6**	**90.0**
16 heads	91.8	86.1
Graph convolution layer types		
GATv2	**92.1**	**87.6**
GraphSAGE	83.7	52.6
GCN	66.2	16.6

Table 4. Ablation study results on the 8-layer architecture with attention and skip connections.

Future Perspectives. While the unified architecture demonstrated robust performance, the observed discrepancy in accuracy between the supply (0.960) and return (0.910) subnets suggests that the current modeling approach may not fully capture the distinct hydraulic regimes. To address this, future work will explore a two-stage extension of the current framework.

First, we will assess the potential benefits of decoupling the networks by training two distinct specialized models. This baseline experiment could help establish the performance ceiling achievable by treating the supply and return domains as separate learning problems. Furthermore, it may be beneficial to implement graph-type conditioning, where the message-passing mechanism explicitly utilizes different weight matrices for supply and return features. Such an approach might allow the model to better distinguish the patterns in data, potentially closing the accuracy gap while maintaining a single deployable system.

4. Conclusions

In this study, we addressed the challenge of fault localization in DHNs under conditions of data scarcity, a common constraint in real-world infrastructure monitoring. By using GNNs, we exploited the topological structure of DHNs to overcome the limitations of traditional physics-based and topology-agnostic data-driven methods.

The research was conducted in two distinct stages. First, we utilized a synthetic dataset to identify an architecture capable of capturing long-range dependencies in graph-structured data. This led to the design of a deep GNN architecture integrating GATv2 node encoders and edge-attention mechanisms. Second, we validated this architecture on a physically modeled dataset generated via thermohydraulic simulations, representing a realistic urban heating network.

The results demonstrated that the proposed model achieves high localization accuracy on the test set: 96.0% for the supply subnet and 91.0% for the return subnet, with corresponding Macro F1 scores of 93.0% and 86.9%, respectively. These metrics were obtained under sparse sensor coverage (only 30 of 187 nodes contained non-zero thermal and pressure features), simulating realistic, data-limited conditions. The ablation studies further confirmed that the attention mechanism and network depth are critical factors for performance. The distance confusion matrix reveals that most errors are localized within a topological distance of 1–2 sections from the actual defect. This confirms that the model implicitly captures the attenuation of hydraulic and thermal disturbances along the network topology.

The current thermohydraulic model is a simplified representation of a real district heating network. It does not include shut-off and control valves (e.g., gate valves, check valves, bypasses), intermediate switching states, or gradual pipe degradation. These factors will be incorporated in future work to reduce the simulation-to-reality gap.

These findings suggest that topology-aware deep learning offers a promising approach for DHN monitoring. Future work will focus on validating the approach on historical field data from operational networks and testing it across different district heating geometries and sensor layouts, while addressing the inevitable domain shift between simulated training regimes and real-world seasonal and load variations.

In practical predictive-maintenance workflows, the proposed GNN can be used as an additional localization and ranking layer. Standard-based risk estimates, pipe age and material data, pipe-static or fatigue assessments, historical water-loss records, leakage-wire alarms, and operating-history data can define prior fault probabilities, while the GNN uses sparse SCADA-like pressure and temperature measurements together with network topology to narrow the inspection area.

Overall, the localization of misclassifications (except for the concentration of errors in segments 32, 36, and 41) supports the idea that high localization quality is achievable with a sufficiently small but well-placed set of sensors, as argued by Rosich et al. [30]. The persistent confusion in this particular region suggests that the current sensor layout lacks sufficient observability there. A possible explanation is that adding a single pressure sensor at the junction of these three segments would eliminate most errors in this region. More generally, the proposed GNN architecture can be used not only for fault localization but also as an analytical tool to guide sensor network augmentation under budget constraints.

However, we recognize that the most significant challenge remains the acquisition of high-quality, labeled data to support such data-driven methodologies.

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Abbreviations

The following abbreviations are used in this manuscript:

CGNN	Composite Graph Neural Network
CNN	Convolutional Neural Network
DHN	District Heating Network
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GNN	Graph Neural Network
GraphSAGE	Graph SAmple and aggreGatE
ReLU	Rectified Linear Unit
SCADA	Supervisory Control and Data Acquisition
SIMPLE	Semi-Implicit Method for Pressure-Linked Equations
SVM	Support Vector Machine
UAV	Unmanned Aerial Vehicle

Nomenclature

A	Cross-sectional area of the pipe
c_p	Specific heat capacity of the fluid
d	Distance metric
e	Graph edge
F	Surface area of the pipe
$f_k(e)$	Target variable function for edge e
k	Number of hops (neighborhood radius)
m_l	Mass flow rate vector along the connected edges
$N_k(e)$	Set of nodes within a k-hop neighborhood of edge e
p	Pressure
p'	Pressure correction
P	Set of edges in the shortest path between nodes
q	Mass source or sink at the specific node
Q	Heat loss from pipe to ambient air
s	Hydraulic resistance coefficient
t_1	Inlet temperature
t_b	Average water temperature within the pipe
t_{out}	Ambient outdoor temperature
t_i	Nodal temperature
u_b	Flow velocity
W_i	Edge feature
X_i	Node attribute
α	Overall heat transfer coefficient
Δp	Pressure drop over the pipe
κ	Coefficient derived from hydraulic resistance $(1/(s\|m_l\|))$
λ	Decay factor
ρ	Fluid density
$\nabla\cdot$	Divergence operator

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